Sec: OSR.IIT_*COSC(MODEL-A&APEX) | GTA-4(P2) | Date: 09-04-26
Time: 3 Hrs | 2024_P2 | Max. Marks: 180

KEY SHEET

MATHEMATICS

1	**D**	2	**A**	3	**B**	4	**B**	5	**AD**
6	**ABCD**	7	**ACD**	8	**4**	9	**1620**	10	**4051**
11	**21**	12	**16**	13	**118**	14	**4043**	15	**0**
16	**2**	17	**2**						

PHYSICS

18	**B**	19	**C**	20	**D**	21	**B**	22	**CD**
23	**BD**	24	**BC**	25	**10**	26	**4**	27	**40**
28	**57**	29	**6**	30	**9**	31	**0.54**	32	**1.25**
33	**1.25**	34	**2**						

CHEMISTRY

35	**B**	36	**C**	37	**A**	38	**B**	39	**AB**
40	**ACD**	41	**CD**	42	**10**	43	**5**	44	**35**
45	**40**	46	**66**	47	**6**	48	**0.40**	49	**2.50**
50	**2**	51	**12**						

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SOLUTIONS
MATHS

1.

As $\omega_1 = \dfrac{2z_2z_3}{z_2+z_3}$, $\omega_2 = \dfrac{2z_1z_3}{z_1+z_3}$, $\omega_3 = \dfrac{2z_1z_2}{z_1+z_2}$

The required equation is $(ab-1)z^3 + 2(b^2+a)z^2 + 8bz + 8 = 0$

2. From the definition we have

$$F'(x) = \frac{x-1}{\ln x}, \quad x > 1.$$

Therefore $F'(x) > 0$ for $x \in (1,\infty)$. Thus F is strictly increasing and hence one-to-one. Since

$$F(x) \geq (x^2-x)\min\left\{\frac{1}{\ln t} \ : \ x \leq t \leq x^2\right\} = \frac{x^2-x}{\ln x^2} \to \infty$$

as $x \to \infty$, it follows that the range of F is $(F(1+),\infty)$. In order to determine $F(1+)$ we substitute $t = e^v$ in the definition of F and we get

$$F(x) = \int_{\ln x}^{2\ln x} \frac{e^v}{v} dv.$$

Hence

$$F(x) < e^{2\ln x}\int_{\ln x}^{2\ln x} \frac{1}{v} dv = x^2 \ln 2$$

and similarly $F(x) > x\ln 2$. Thus $F(1+) = \ln 2$.

3.

Solution. Put $a_n = 1 + \sqrt{b_n}$ for $n \geq 0$. Then $a_n > 1$, $a_0 = 2$ and

$$a_n = 1 + \sqrt{1 + a_{n-1} - 2\sqrt{a_{n-1}}} = \sqrt{a_{n-1}},$$

so $a_n = 2^{2^{-n}}$. Then

$$\begin{aligned}\sum_{n=1}^{N} b_n 2^n &= \sum_{n=1}^{N}(a_n-1)^2 2^n = \sum_{n=1}^{N}[a_n^2 2^n - a_n 2^{n+1} + 2^n] \\ &= \sum_{n=1}^{N}[(a_{n-1}-1)2^n - (a_n-1)2^{n+1}] \\ &= (a_0-1)2^1 - (a_N-1)2^{N+1} = 2 - 2\frac{2^{2^{-N}}-1}{2^{-N}}.\end{aligned}$$

Put $x = 2^{-N}$. Then $x \to 0$ as $N \to \infty$ and so

$$\sum_{n=1}^{\infty} b_n 2^N = \lim_{N\to\infty}\left(2 - 2\frac{2^{2^{-N}}-1}{2^{-N}}\right) = \lim_{x\to 0}\left(2 - 2\frac{2^x-1}{x}\right) = 2 - 2\ln 2.$$

4.

The normal vector of the plane is $n = \begin{vmatrix} i & j & k \\ 1 & 0 & 0 \\ 2 & 1 & 3 \end{vmatrix} = 3j - k$ and it passes through $(-1,1,-2)$.

Thus its equation is $3y - z = 5$. The vector v which makes the least angle with u is along the projection of u on the plane.

$n_1 = 2i + j + 3k, n_2 = k$, then the direction of L_2 is $n_1 \times (n_1 \times n_2) = j + 3k$.

Let $\overrightarrow{CQ} = \lambda(j+3k)$. Since $CQ = \sqrt{10} \Rightarrow \lambda = \pm 1$.

Thus $Q = (0,-1,-8)$

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5.

Since, $f'(x) > 0 \Rightarrow f(x)$ is increasing function

$\Rightarrow f(0) = 0, f(1) = 1$

Let $g(x) = \frac{f(x)}{x}$, $x \in (0, 1] \Rightarrow g'(x) = \frac{xf'(x) - f(x)}{x^2}$

Let $p(x) = xf'(x) - f(x)$; $x \in [0, 1]$

$\Rightarrow p'(x) = xf''(x) + f'(x) - f'(x) < 0 \ \forall \ x \in (0, 1]$

$\Rightarrow p(x) < p(0) = -f(0) = 0 \Rightarrow g'(x) < 0 \ \forall \ x \in (0, 1]$

$\Rightarrow \frac{f(x)}{x} \geq \frac{f(1)}{1} \Rightarrow \frac{f(x)}{x} \geq 1 \Rightarrow f(x) \geq x \Rightarrow x \geq f^{-1}(x)$

6.

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots\ldots \quad \ldots\ldots (1)$$

$$\omega^2 \ln(1+\omega x) = x - \frac{\omega x^2}{2} + \frac{\omega^2 x^3}{3} \ldots\ldots \quad \ldots\ldots (2)$$

$$\omega \ln(1+\omega^2 x) = x - \frac{\omega^2 x^2}{2} + \frac{\omega x^3}{3} - \frac{x^4}{4} + \ldots\ldots \quad \ldots\ldots (3)$$

Adding equation (1) + (2) + (3), we get $S = \frac{1}{3}\left(\ln 2 + \frac{\pi}{\sqrt{3}}\right)$

7. $\int_0^{\frac{\pi}{2}} \log(\sin x)\,dx = -\frac{\pi}{2}\log 2$

$$F(y) = \int_0^1 \frac{\tan^{-1}xy}{x\sqrt{1-x^2}}dx \Rightarrow F'(x) = \int_0^1 \frac{dx}{(1+x^2y^2)\sqrt{1-x^2}} = \int_0^{\frac{\pi}{2}} \frac{dx}{1+y^2\cos^2 t}$$

$$= \frac{1}{\sqrt{1+y^2}}\tan^{-1}\left(\frac{tant}{\sqrt{1+y^2}}\right) = \frac{\pi}{2\sqrt{1+y^2}} \Rightarrow$$

$$F(y) = \frac{\pi}{2}log\left(y + \sqrt{1+y^2}\right) + C$$

$$\int_0^1 \frac{\tan^{-1}x}{x\sqrt{1-x^2}}dx = \frac{\pi}{2}log\left(1+\sqrt{2}\right); (2)$$

8. Given $\frac{a_0}{1-r} = 3$ where, r is the ratio of the geometric sequence

Required sum $S_\infty = \frac{a_0}{1-r^3}$

Dividing $S_\infty = \frac{3}{1+r+r^2}$ *for* $|r| < 1$

Hence S_∞ is maximum at $r = \frac{-1}{2}$ and $S_\infty = 4$

9. graph of $\frac{x^2}{100} + \frac{y^2}{19} = 1$ will be an ellipse having a horizontal major axis with length $2\sqrt{100} = 20$ and vertical minor axis with length $2\sqrt{19}$. The foci are at $(\pm c, 0)$, where $c^2 = 100 - 19 = 81$, and hence $c = 9$. In other words, the foci are a distance of 18 apart, so when the paper is rolled up into a cylinder the circular cross-section will have circumference 18. But this means its radius r satisfies $2\pi r = 18$, so $r = 9/\pi$. Therefore its area equals $\pi r^2 = 81/\pi$, which results in a volume of $20 \cdot (81/\pi) = \mathbf{1620}/\pi$.

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10. $$\frac{2n^2-1}{n^2(n+1)^2}=\frac{An+B}{n^2}+\frac{Cn+D}{(n+1)^2}=\frac{2n-1}{n^2}+\frac{-2n-1}{(n+1)^2}=\frac{2n-1}{n^2}-\frac{2(n+1)-1}{(n+1)^2}$$

$$\sum_{n=1}^{2025}\frac{2n^2-1}{n^2(n+1)^2}=\sum_{n=1}^{2025}\left(\frac{2n-1}{n^2}-\frac{2(n+1)-1}{(n+1)^2}\right)=\frac{2\cdot1-1}{1^2}-\frac{2(2025+1)-1}{(2025+1)^2}$$

$$=\frac{2026^2-2\cdot2026+1}{2026^2}=\left(\frac{2025}{2026}\right)^2$$

Taking the square root of this expression, we get the answer $\boxed{\frac{2025}{2026}}$.

11. $$RHS=\frac{\left(\sqrt{3}+i\right)^{40}+\left(\sqrt{3}-i\right)^{40}}{2}=(-2)^{39}=(-8)^{13}$$

12.

Equation of the normal at 'α' on the ellipse is $\frac{2x}{\cos\alpha}-\frac{y}{\sin\alpha}=3$

IF it passes through 'θ', then $\frac{4\cos\theta}{\cos\alpha}-\frac{\sin\theta}{\sin\alpha}=3$

$$\Rightarrow\frac{16\cos^2\theta}{\cos^2\alpha}=\left(3+\frac{\sin\theta}{\sin\alpha}\right)^2$$

$$\Rightarrow\sin^2\theta\left(\frac{16}{\cos^2\alpha}+\frac{1}{\sin^2\alpha}\right)+\frac{6\sin\theta}{\sin\alpha}+9-\frac{16}{\cos^2\alpha}=0$$

$$\therefore\sin\alpha\sin\beta=\sin^2\alpha\left(\frac{9\cos^2\alpha-16}{16\sin^2\alpha+\cos^2\alpha}\right)$$

$$\Rightarrow\sin\beta=f(\alpha)=-\sin\alpha\left(\frac{7+9\sin2\alpha}{1+15\sin^2\alpha}\right)$$

13.

$P(HH)=p^2$ $P(TT)=q^2$ $P(TH \text{ or } HT)=2pq$

P(first $n-1$ rounds indecisive) $=(2pq)^{n-1}$

$\Rightarrow$ Decision at round $n=(2pq)^{n-1}\times$ P(HH or TT)

$=(2pq)^{n-1}(p^2+q^2)$

14&15.

$$\left(\lim_{n\to\infty}\sum_{i=1}^{n}x_i^{2021}f(x_i)\Delta x\right)^2\le\left(\lim_{n\to\infty}\sum_{i=1}^{n}f^2(x_i)\Delta x\right)\left(\lim_{n\to\infty}\sum_{i=1}^{n}x_i^{4042}\Delta x\right)$$

$$=\int_0^1 f^2(x)dx\int_0^1 x^{4042}dx=\frac{1}{4043}$$

16&17. When the graphs of ae^x and x^3 are tangent to each other, i.e, at x_0 such that $ae^{x_0}=x_0^3$ and $ae^{x_0}=3x_0^2$. From these two equations we get $x_0=3$ and $a=\frac{27}{e^3}$. Summarizing: the equation $ae^x=x^3$ has one solution for $a\le0$ and $a=\frac{27}{e^3}$, two solutions for $0<a<\frac{27}{e^3}$ and no solutions for $a>\frac{27}{e^3}$.

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18. Charge will perform an helical path with non-uniform pitch. Such that

$\frac{2v_2}{\left(qE/m\right)} = n\left(\frac{2\pi m}{qB}\right)$; n is a positive integer

19. The entire system acts as silvered lens aka mirror of focal length f, such that

$$\frac{1}{f} = \frac{2}{-10} - 2\left[\left(\frac{1.8}{1}-1\right)\left(\frac{1}{-40}-\frac{1}{-20}\right)+\left(\frac{1.1}{1}-1\right)\left(\frac{1}{-20}-\frac{1}{-10}\right)\right] = \frac{-1}{4}$$

$$\frac{1}{v}+\frac{1}{-12} = \frac{1}{-4} \Rightarrow v = -6cm$$

20. $$\vec{F} = \frac{\left[\alpha\left|\vec{r}-\vec{b}\right|^3+\beta\left|\vec{r}-\vec{a}\right|^3\right]\vec{r}-\left[\alpha\vec{a}\left|\vec{r}-\vec{b}\right|^3+\beta\vec{b}\left|\vec{r}-\vec{a}\right|^3\right]}{\left[\left|\vec{r}-\vec{a}\right|\left|\vec{r}-\vec{b}\right|\right]^3} = \frac{\alpha\left(\vec{r}-\vec{a}\right)}{\left|\vec{r}-\vec{a}\right|^3}+\frac{\beta\left(\vec{r}-\vec{b}\right)}{\left|\vec{r}-\vec{b}\right|^3}$$

That simplified expression of force is equivalent to a positive point charge moving under the influence of electrostatic repulsion of two other fixed positive point charges. Therefore there is one unstable equilibrium the two fixed charges.

21. ρ fluid applies a force F downwards such that F and the force at the cross-section of ρ-2ρ interface put together form the buoyant force of ρ

$$P_1\left(\pi H^2\right)-F = B_1 \Rightarrow F = \rho g\left(2H\right)\left(\pi H^2\right)-\rho\left(\frac{1}{3}\pi H^2 H\right)g \Rightarrow F = \frac{5}{3}\pi\rho g H^3$$

Similarly, force of 3ρ and the force at the cross-section of 2ρ-6ρ interface put together form the buoyant force of 3ρ

$$F_2 - P_2\left(\pi\left(3H\right)^2\right) = B_3$$

$$\Rightarrow F_2 = \left[\rho g\left(2H\right)+\left(2\rho\right)g\left(2H\right)\right]\left(9\pi H^2\right)+6\rho\left(\frac{1}{3}\pi\left(4H\right)^2\left(4H\right)-\frac{1}{3}\pi\left(3H\right)^2\left(3H\right)\right)g$$

$$\Rightarrow F_2 = 128\pi\rho g H^3$$

22. Once the key is closed let a charge x circulate in clockwise sense, then

$$\frac{2q-x}{C}-\frac{q+x}{2C}-\frac{x}{C} = 0 \Rightarrow x = \frac{3}{5}q$$

New charge on $C_1 = q+\frac{3}{5}q = \frac{8}{5}q$, Initial energy in C_1 was $\frac{q^2}{4C}$ and new energy is $\frac{16q^2}{25C}$, therefore gain in energy by C_1 is $\frac{39q^2}{100C}$.

New charge on $C_2 = 2q-\frac{3}{5}q = \frac{7}{5}q$, Initial energy in C_2 was $\frac{2q^2}{C}$ and new energy is $\frac{49q^2}{50C}$, therefore loss in energy by C_2 is $\frac{51q^2}{50C}$.

New charge on $C_3 = \frac{3}{5}q$, there was no Initial energy in C_3, whereas after closing the key energy is $\frac{9q^2}{50C}$, therefore gain in energy by C_3 is $\frac{9q^2}{50C}$.

Heat generated during the closing of the key is $\frac{51q^2}{50C}-\frac{39q^2}{100C}-\frac{9q^2}{50C} = \frac{9q^2}{20C}$

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23. $FR=\left(I_{cm}+mR^2\right)\alpha$

$f_sR=I_{cm}\alpha$

$a=R\alpha$

Solving the above three equations we can say that: $a=\dfrac{f_sR^2}{I_{cm}}=\dfrac{FR^2}{\left(I_{cm}+mR^2\right)}$

24. If O is node: $f=\dfrac{pv_1}{2(5l)}=\dfrac{qv_2}{2l}\Rightarrow\dfrac{p}{2(5l)}\sqrt{\dfrac{F}{\mu}}=\dfrac{q}{2l}\sqrt{\dfrac{F}{9\mu}}\Rightarrow\dfrac{p}{5}=\dfrac{q}{3}\Rightarrow p=5n,q=3n$

$f=\dfrac{5nv_1}{2(5l)}=\dfrac{3nv_2}{2l}\Rightarrow\lambda_1=\dfrac{v_1}{f}=\dfrac{2l}{n},\lambda_2=\dfrac{v_2}{f}=\dfrac{2l}{3n}$

As O, P and Q are nodes: OP $=\alpha\dfrac{\lambda_1}{2}=\dfrac{\alpha l}{n}$ and OQ $=\beta\dfrac{\lambda_2}{2}=\dfrac{\beta l}{3n}$

Therefore, PQ $=\alpha\dfrac{\lambda_1}{2}+\beta\dfrac{\lambda_2}{2}=\dfrac{\alpha l}{n}+\dfrac{\beta l}{3n}=\dfrac{l}{n}\left(\alpha+\dfrac{\beta}{3}\right)=\dfrac{l}{3}\left(\dfrac{3\alpha+\beta}{n}\right)$

If O is Anti-node: $f=\dfrac{(2p+1)v_1}{4(5l)}=\dfrac{(2q+1)v_2}{4l}\Rightarrow\dfrac{2p+1}{5}=\dfrac{2q+1}{3}\Rightarrow p=5n+2,q=3n+1$

$f=\dfrac{5(2n+1)v_1}{4(5l)}=\dfrac{3(2n+1)v_2}{4l}\Rightarrow\lambda_1=\dfrac{v_1}{f}=\dfrac{4l}{2n+1},\lambda_2=\dfrac{v_2}{f}=\dfrac{4l}{3(2n+1)}$

As O is Anti-node, P and Q are nodes: OP $=(2\alpha+1)\dfrac{\lambda_1}{4}=\dfrac{(2\alpha+1)}{2n+1}l$ and OQ

$=(2\beta+1)\dfrac{\lambda_2}{4}=\dfrac{(2\beta+1)}{3(2n+1)}l$

Therefore, PQ $=\dfrac{(2\alpha+1)}{2n+1}l+\dfrac{(2\beta+1)}{3(2n+1)}l=\dfrac{l}{3}\left(\dfrac{6\alpha+2\beta+4}{2n+1}\right)$

25. For Vernier Calipers:
1 main scale division = 1/8cm
1 vernier scale division = 1/10cm
So least count = 1/40cm
For screw gauge:
Pitch (p) = 2 main scale division = 2(2 X 1/40)cm = 1/10cm
So lest count = p/100 = 1/1000cm = 10μm

26. Let the resistances be x, y and z.

When S_1 is closed: $\dfrac{V^2}{x+y}=1\Rightarrow x+y=V^2$

When S_1 and S_3 are closed: $\dfrac{V^2}{x+\dfrac{yz}{y+z}}=2\Rightarrow\dfrac{xy+yz+zx}{y+z}=\dfrac{V^2}{2}$

When S_4 alone is closed: $\dfrac{V^2}{y+z}=2\Rightarrow y+z=\dfrac{V^2}{2}$

$\Rightarrow xy+yz+zx=\dfrac{V^4}{4}$

When S_2 and S_4 are closed: $P=\dfrac{V^2}{z+\dfrac{xy}{x+y}}=\dfrac{(x+y)V^2}{xy+yz+zx}=\dfrac{V^2\times V^2}{\dfrac{V^4}{4}}=4$

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27. $\frac{2\pi}{\lambda}\left[\left(SS_2+S_2P\right)-\left(SS_1+S_1P\right)\right]=2n\pi \Rightarrow \left(SS_2-SS_1\right)+d\sin\theta=n\lambda$

$\Rightarrow (174-121.1)\times10^{-6}+10\times10^{-6}\sin\theta=n\left(5\times10^{-7}\right)\Rightarrow 5.29+\sin\theta=\frac{5n}{100}$

$\because -1\le\sin\theta\le1\Rightarrow -4.29\le5.29+\sin\theta\le6.29\Rightarrow 4.29\le\frac{5n}{100}\le6.29\Rightarrow 85.8\le n\le125.8$

Therefore, number of ns is -86 + 1 + 125 = 40

28. Mosley's Law: $\sqrt{f_{K_\alpha}}=a(Z-1)\Rightarrow\sqrt{\frac{c}{\lambda_{K_\alpha}}}=a(Z-1)\Rightarrow\sqrt{\frac{\lambda_1}{\lambda_2}}=\frac{Z_2-1}{Z_1-1}$

$\sqrt{\frac{1.5404}{0.3851}}=\frac{Z-1}{29-1}\Rightarrow\frac{Z-1}{28}=\sqrt{4}\Rightarrow Z=57$

29. F = ILB, for F vs L slope S is IB

S = IB $\Rightarrow B=\frac{S}{I}\Rightarrow\frac{\Delta B}{B}=\frac{\Delta S}{S}+\frac{\Delta I}{I}=\frac{1}{25}+\frac{1}{50}=\frac{3}{50}=\frac{6}{100}$

30. When switches are closed each branch between G_1 and G_2 will have zero potential difference. Therefore, charge on each capacitor is $\varepsilon_iC_i\ \forall i\in\{1,2,3\}$. Charge into ground G_1 is

$\varepsilon(3C)+(2\varepsilon)(6C)-(3\varepsilon)(2C)=9\varepsilon C$

31. The object will move in an elliptical path with the centre O of the planet as one of the foci. As the velocities at P and Q are antiparallel, these point must be the semi-minor point of the ellipse and their mid- point C must be the centre of the ellipse.

$OC=R\cos\theta$, $PQ=2PC=2R\sin\theta$, Semi-major length $a=R$, Semi-minor length $b=PC=R\sin\theta$

Area of triangle POQ = $\frac{1}{2}(OC)(PQ)=R^2\sin\theta\cos\theta$

From Kepler's 2nd law, we can say:

$\frac{\Delta t}{T}=\frac{Area\ of\ \Delta POQ+\frac{\pi ab}{2}}{\pi ab}=\frac{R^2\sin\theta\cos\theta}{\pi R(R\sin\theta)}+\frac{1}{2}=\frac{\cos\theta}{\pi}+\frac{1}{2}$, where $T=\frac{2\pi}{\sqrt{GM}}R^{3/2}$

32. Kinetic energy is minimum at the maximum height point H. Then, OH = OC + CH = $R\cos\theta+a$

OH = $R\cos\theta+R=R(1+\cos\theta)$

$U_H=\frac{-GMm}{R(1+\cos\theta)}$ and we know that the total energy of an elliptical path is $E=\frac{-GMm}{2a}=\frac{-GMm}{2R}$

$K_{\min}+U_H=E\Rightarrow K_{\min}=\frac{-GMm}{2R}-\frac{-GMm}{R(1+\cos\theta)}$

$\Rightarrow K_{\min}=\frac{GMm}{2R}\left[\frac{2}{(1+\cos\theta)}-1\right]\Rightarrow\frac{GMm}{2RK_{\min}}=\frac{1+\cos\theta}{1-\cos\theta}$

33. Let the resistance and reactance of Z_1 be R and X, then

$\frac{R}{Z_1}=0.6\Rightarrow R=\frac{3}{5}Z_1\Rightarrow R=\frac{3}{5}\sqrt{R^2+X^2}\Rightarrow16R^2=9X^2\Rightarrow X=\pm\frac{4}{3}R$

and $\frac{R}{\sqrt{R^2+(L\omega+X)^2}}=0.8\Rightarrow\frac{R}{\sqrt{R^2+\left(L\omega\pm\frac{4}{3}R\right)^2}}=\frac{4}{5}$

On solving we get $R=\frac{12}{25}L\omega$ or $\frac{12}{7}L\omega$

$\Rightarrow\frac{3}{5}Z_1=\frac{12}{25}L\omega$ or $\frac{12}{7}L\omega\Rightarrow Z_1=\frac{4}{5}L\omega$ or $\frac{20}{7}L\omega$

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34. Let the resistance and reactance of Z_2 be R and X, then

$$\frac{R}{Z_2}=0.8\Rightarrow R=\frac{4}{5}Z_4\Rightarrow R=\frac{4}{5}\sqrt{R^2+X^2}\Rightarrow 9R^2=16X^2\Rightarrow X=\pm\frac{3}{4}R$$

and $$\frac{R}{\sqrt{R^2+\left(L\omega+X\right)^2}}=0.6\Rightarrow\frac{R}{\sqrt{R^2+\left(L\omega\pm\frac{3}{4}R\right)^2}}=\frac{3}{5}$$

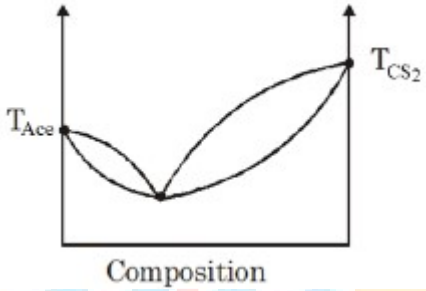
On solving we get $R=\frac{12}{25}L\omega$ *or* $\frac{12}{7}L\omega$

$$\Rightarrow\frac{4}{5}Z_2=\frac{12}{25}L\omega \text{ or } \frac{12}{7}L\omega\Rightarrow Z_2=\frac{3}{5}L\omega \text{ or } \frac{15}{7}L\omega$$

$$\Rightarrow Z_2=\frac{15}{7}L\omega\Rightarrow\frac{7Z_2}{15L}=2\pi f$$

CHEMISTRY

35.



36. Na/ C_2H_5OH cannot reduce Carboxylic acid.

39. a) $Cu(OH)_2$ and $Ni(OH)_2$ are Precipitated
b) $[Cu(NH3)_4]^{2+}$ $[Ni(NH3)_6]^{2+}$ are formed and Zinc is Precipitated as $Zn(OH)_2$
c) $K3[Cu(CN)_4]$ is formed
d) $[Ni(NH3)_6]^{2+}$ $[Zn(NH3)_4]^{2+}$ $[Cu(NH3)_4]^{2+}$

41. B) $Nu^{\odot}$ Addition is effected due to bulky groups

C) Ozonolysis followed by Aldol condensation.

D) Aldol Condensation

42. White PPt is Cu_2I_2 spin only Magnetic Moment : 0

Other Metal Ion in Solution X is Zinc. Total electron in d- subshell: 10

43. Ag^+ displaced to interstitial sites

$$=N\,X\,\frac{2}{100}=0.02\,N$$

Ag^+ missing due to schottky defect

$$=N\,X\,\frac{0.5}{100}=0.05\,N$$

Br^- missing due to schottky defect

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$= N\,X\,\frac{0.5}{100} = 0.05\,N$

No .of Ag^+ at lattice sites

$= N - 0.2N - 0.005N = 0.975N$

Total number of $Br^- = N - 0.005N = 0.995N$

$\frac{\rho}{\rho_0} = \frac{mass\ of\ defective\ crystal}{mass\ of\ perfect\ crystal}$

$= \frac{0.995\,N}{N} = 0.995$

Total number of tetrahedral voids = 2N

Fraction of tetrahedral voids occupied $= \frac{0.02N}{2N} = 0.01$

Co-ordination number of tetrahedral void = 4

44.

$Co^{2+} + 6NH_3 \rightleftharpoons \left[Co(NH_3)_6\right]^{2+} \quad K_1 = 10^5$

$Co^{3+} + 6NH_3 \rightleftharpoons \left[Co(NH_3)_6\right]^{3+} \quad K_2 = 10^x$

$Co^{3+} + e \rightarrow Co^{2+} \quad E^o = 1.90\,V$

$\left[Co(NH_3)_6\right]^{2+} \rightarrow \left[Co(NH_3)_6\right]^{3+} + e \quad E^o = -0.1V$

$Co^{3+} + \left[Co(NH_3)_6\right]^{2+} \rightleftharpoons Co^{2+} + \left[Co(NH_3)_6\right]^{3+} \quad E^0 = 1.8V$

$E^o = \frac{0.06}{1}\log\frac{K_2}{K_1}(at\ equilibrium)$

$1.8 = 0.06\log\frac{K_2}{10^5}$

$30 = \log\frac{K_2}{10^5}$

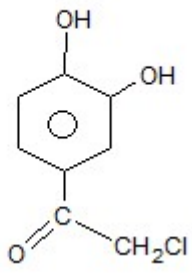
$K_2 = 10^{35}$

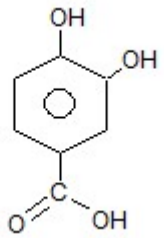
45. $HSO_4^{1-} \rightarrow H_2S_2O8 \xrightarrow{KOH} K_2S_2O_8 \xrightarrow{Mn^{+2}} MnO_4^{1-}$

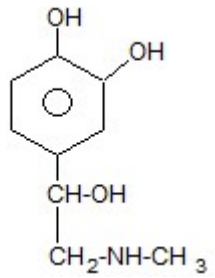
$8MnO_4^{1-} + 3S_2O_3^{2-} + H_2O \rightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^-$

46. $Z = SnCl_4 \quad X = Cl - CH_2 - CH_2 - S - CH_2 - CH_2 - Cl \quad Y = Cu_3P_2$

47. All statements are correct except iii

X = 

Y = 

Z = 

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48 & 49.

$$NO(g) + NO_2(g) \rightleftharpoons N_2O_3(g)$$

$3P \qquad 5P \qquad 0$

$3P - X \qquad 5P - X - 2y \qquad X$

$$2NO_2(g) \rightleftharpoons N_2O_4(g)$$

$5P - X - 2y \qquad y$

$$K_{P_2} = \frac{P_{N_2O_4}}{(P_{NO_2})}, \qquad 8 = \frac{P_{N_2O_4}}{(0.5)^2}$$

$P_{N_2O_4} = 2 \ atm$

$\therefore y = 2 atm$

$P_{total} = P_{NO} + P_{NO_2} + P_{N_2O_3} + P_{N_2O_4}$

$5.5 = 3P - X + 0.5 + X + 2$

$3P = 3, P = 1$

$5P - x - 2y = 0.5$

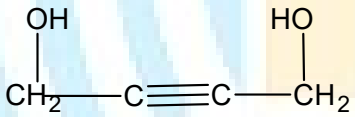
$5 \times 1 - X - 2 \times 2 = 0.5, X = 0.5 \ atm$

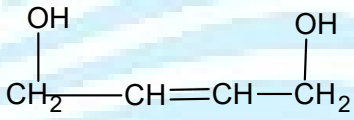
$$K_{P_1} = \frac{P_{N_2O_3}}{(P_{NO})(P_{NO_2})}$$

$$= \frac{0.5}{2.5 \times 0.5} = 0.4 \ atm$$

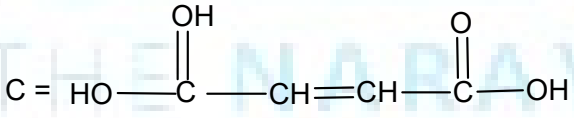
$P_{NO} = 2.5 \ atm \ P_{NO_2} = 0.5 \ atm \ P_{N_2O_3} = 0.5 \ atm, P_{N_2O_4} = 2 \ atm$

51. R = 7 S= 5

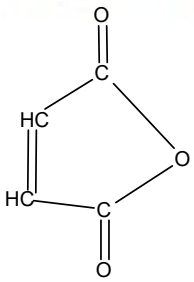
A = 

B = 

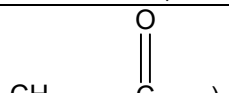
cis Isomer

C = 

cis Isomer

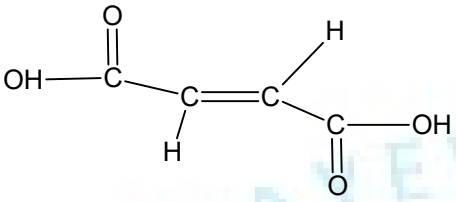
D = 

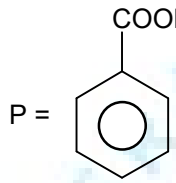
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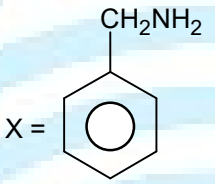
$E = A_cO - CH_2 - C \equiv C - CH_2 - OA_c$ (Ac = )

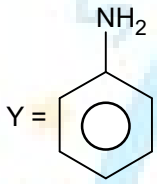
$F = A_cOH_2C - CH = CH - CH_2OAc$ $(Trans\ Isomer)$

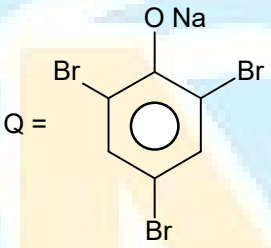
$G = HOH_2C - CH = CH - CH_2OH$ $(Trans\ Isomer)$

H = 

P = 

X = 

Y = 

Q = 

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